

OPAC-018 Green-Schur Bridge: External-Review Package

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Abstract

This manuscript source accompanies an external-review research package for OPAC-018. The package studies the root-polytope projection bound $\kappa(\Phi, U) < 2$ through documented rational formulas, exact arithmetic audits, negative controls, and explicit open audit items. It is not presented as an independently accepted theorem; external mathematical review remains pending.

1 Truth Boundary

The repository label is

SOLUTION-CANDIDATE / EXTERNAL REVIEW PENDING.

The software verifies encoded formulas and conservative envelopes over exact rational arithmetic. It does not replace independent mathematical review of the reductions that produce those formulas.

2 Basic Setup

Let Φ be a finite classical root system in a Euclidean vector space. Let

$$P_\Phi = \text{conv}(\Phi)$$

be the root polytope. For a root-spanned subspace U , let π_U denote orthogonal projection and let $\Psi = \Phi \cap U$. The target containment is encoded by the gauge

$$\gamma_\Psi(y) = \inf\{\lambda > 0 : y \in \lambda P_\Psi\}.$$

The dilation factor is reviewed through

$$\kappa(\Phi, U) = \max_{\alpha \in \Phi} \gamma_\Psi(\pi_U(\alpha)).$$

3 Representative Formulas Audited by the Package

For Type A two-block configurations with finite block sizes $b_1, b_2 \geq 2$, the encoded formula is

$$\kappa_A(b_1, b_2) = 2 - \frac{1}{b_1} - \frac{1}{b_2}.$$

For Type C single-block configurations with $b_1 \geq 2$, the encoded envelope is

$$\kappa_C(b_1) = \max\left(1, 2 - \frac{2}{b_1}\right).$$

The software layer also includes conservative Type B and Type D encoded envelopes under the assumptions documented in the repository. Reviewers should inspect the Type B and Type D mathematical reductions separately.

4 Exact Arithmetic Audit Philosophy

All proof-critical comparisons use rational arithmetic. A comparison

$$\frac{a}{b} < \frac{c}{d}$$

with positive denominators is checked by the integer cross-product

$$ad < cb.$$

Floating-point values may be printed for display only, but they are not used in the boundary decision.

5 Reviewer Navigation

Start with `START_REVIEW_HERE.md`. Then inspect `DEPENDENCY_TABLE.md`, `docs/FORMAL_ASSUMPTIONS.md`, the worked examples in `examples/`, and this manuscript source. The strongest review should verify the gauge identity scope, signed-graph structural coverage, Type B and Type D case split, and exceptional-family exclusion boundary.

6 Reproducibility Commands

```
python -m compileall -q .
python pure_python_exact_audit.py
python counterexample_stress_test.py
python verify_all.py
pytest -q
python verify_manifest.py
python build_sha_manifest.py --check
```

7 Conclusion

This release packages the Shawn Calvin Snelling / AXEZENT AI OPAC-018 Green-Schur Bridge work as a review-ready solution-candidate repository with deterministic exact-audit infrastructure. Independent mathematical validation remains the required next step before any final theorem-status claim.